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Nurse Work Environment and Hospital Readmission Disparities Between Patients With and Without Limited English Proficiency

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ABSTRACT

Minimal progress has been made in narrowing disparities between patients with and without limited English proficiency (LEP). Using 2016 data from RN4CAST-US, New Jersey Discharge Data Collection System, and AHA Annual Hospital Survey, multivariable logistic regression models were employed to examine whether and to what extent the hospital nurse work environment, defined as the conditions that nurses work in, is associated with decreased disparities in 7-day hospital readmissions between patients with and without LEP. Existing literature has established associations between nurse work environments and outcomes disparities of various minoritized populations; however, no literature has examined this relationship in the context of hospitalized patients with LEP. In a sample of 424,745 hospitalized adults ($n = 38{,}906$ with LEP), patients with LEP, compared to those without LEP, were younger (63.4 vs 64 years old, $p < 0.001$), more likely to be insured by Medicaid (8.9% vs 5.5%) or uninsured (7.5% vs 2%, $p < 0.001$), and readmitted (4.5% vs 3.9%, $p < 0.001$). Adjusting for patient and hospital characteristics, LEP patients had 33% higher odds of a 7-day readmission, as compared to patients without LEP (OR 1.33, 95% CI [1.19–1.47]). A significant interaction was found between patients' LEP status and the nurse work environment (OR 0.83, 95% CI [0.70–0.99]), such that patients with LEP experienced lower odds of 7-day readmission in more favorable nurse work environments, compared to patients without LEP. Hospitals dedicated to providing equitable healthcare may consider enhancing nurses' working conditions as a potential way to reduce disparities in readmission rates.

PATIENT OR PUBLIC CONTRIBUTION

This cross-sectional study leverages three merged secondary data sources from 2016, including data from the New Jersey Discharge Data Collection System (NJDDCS) which contains inpatient hospital discharge encounters from patients in New Jersey and provides patient-level demographic data and clinical information pertaining to patients' hospitalization. All patient-level data used in this study were de-identified

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1 | Introduction

For persons with limited English proficiency (LEP n.d.) in the United States, language barriers impact virtually every aspect of life. When navigating a complex system like health care, the impact is even greater because of how it affects care delivery. High-quality communication is essential for delivering high-quality and safe healthcare, especially when clinicians conduct physical assessments, discuss diagnoses, and explain medication plans.

Disparities in health outcomes for patients with LEP are well-established, especially in acute care settings. Compared to English-proficient individuals, hospitalized patients with LEP are more likely to experience higher rates of hospital readmissions, (Squires et al. 2022), adverse events (Divi et al. 2007; van Rosse et al. 2016), and increased risk of in-hospital mortality (Castro et al. 2022). Although using language services (e.g., interpreters) is the gold standard of care when caring for patients with LEP, language services are poorly utilized in hospital care (Kale and Syed 2010; Taira and Orue 2019). Individuals with LEP are legally entitled to an interpreter under Title VI of the U.S. Civil Rights Act 2023. Section 1557 of the Affordable Care Act, which took effect on July 1, 2016, further clarified the requirements for language services in healthcare and mandates that language services be provided at no cost, be delivered by qualified interpreters, and include taglines in the top 15 non-English languages in the state on all significant documents. This law also requires hospitals to certify staff members who claim proficiency in another language and prohibits staff from interpreting unless they have received formal interpreter training (Squires and Youdelman 2019). Importantly, this study can serve as a baseline for a future policy evaluation to determine whether Section 1557 of the ACA resulted in a substantial reduction in disparities in care quality and costs, both of which were initial objectives of the ACA (Silvers 2013).

During a hospitalization, nurses are the personnel most likely to use language services with the greatest frequency because they are responsible for round-the-clock care and its associated activities. Theoretically, the quality of nurses' work environments may contribute to the underutilization of language services or other important care provisions that contribute to equitable outcomes. For example, a hospital with poor staffing ratios may result in nurses having limited time to obtain, setup, and use these language services (Betancourt et al. 2012; Gerchow et al. 2020). Inadequate staffing ratios and nurses having time constraints are common themes in qualitative literature and suggest that the conditions that nurses work in—known as the work environment—may impact outcomes for patients with LEP (Feiring and Westdahl 2020; Gerchow et al. 2020; Squires et al. 2019).

Existing literature demonstrates that when nurses are supported in their hospital work environment with safe staffing resources, foundations for high-quality patient care delivery, autonomy in unit decision-making with supportive nurse management and collegial nurse-physician relations, patients experience better outcomes such as reduced readmissions for Hispanic surgical patients (Daus et al. 2024) and lower emergency department revisits for patients with LEP (Muir et al. 2025). Minimal research exists about whether these same attributes of the nurse work environment are important for reducing readmission disparities among hospitalized patients with LEP. The aim of this novel study is to determine whether and to what extent the hospital nurse work environment is associated with readmission disparities between patients with and without LEP. We hypothesized that 7-day readmission disparities between patients with and without LEP would be narrower in hospitals with more favorable nurse work environments, compared to hospitals with poor nurse work environments. The short time between hospitalization and readmission suggests that 7-day readmissions may serve as an indicator of the quality of care of the initial hospitalization (e.g., failure to provide an accurate diagnosis, perform comprehensive patient education, or make timely connections to resources related to housing, food security, or transportation). These readmissions are often used as a measure of care quality (e.g., Hospital Readmission Reduction Program) and can have financial implications for hospitals (Chin et al. 2016; McIlvennan et al. 2015).

2 | Methods

2.1 | Study Design and Data

This study used a cross-sectional design leveraging three merged data sources from 2016 when Section 1557 of the ACA was initiated; thus, the analysis provides foundational data to examine the relationship between hospital nursing work environments and patients with LEP's outcomes before policy implementation. The data set created for this study included: (1) Patient hospitalization records from the New Jersey Discharge Data Collection System (NJDDCS); (2) Data about hospital nursing resources from the RN4CAST-US survey of nurses; and (3) Data of hospital characteristics from the American Hospital Association (AHA) Annual Survey.

The NJDDCS contains inpatient hospital discharge records from all patients across New Jersey hospitals and provides patient-level demographic data and clinical information pertaining to patients' hospitalization. The NJDDCS is unique compared to other state discharge datasets because it includes patient language preference. The RN4CAST-US survey offers valuable data from a random 30% sample of registered nurses regarding their working conditions, including questions related to their work environments. Nurses are asked to provide the name of their employer, enabling the aggregation of survey responses at the hospital level to analyze nursing organizational factors. Details about the survey methodology have been published elsewhere (Lasater et al. 2019; Sloane et al. 2018).

2.2 | Study Sample

The patient sample consisted of 424,745 hospitalized adults (aged 18 to 100) who were discharged from one of 68 general acute care hospitals in New Jersey for medical or surgical reasons between January 1, 2016, and December 31, 2016. Patients who had an initial discharge date later than December 26, 2016, died in the hospital, were discharged to hospice, left against medical advice, were admitted for psychiatric reasons, drug or alcohol, or labor, and delivery-related conditions, or those with lengths of stay greater than 60 days were excluded from the

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TABLE 1 | Characteristics and readmission outcomes of patients with and without limited English proficiency ($n = 424{,}745$)[a].

	All ($n = 424{,}745$)	LEP ($n = 38{,}906$)	Non-LEP ($n = 385{,}839$)	Effect Size	*p* value
Patient Characteristics					
Age, mean (SD)	64 (18.3)	63.4 (18.1)	64 (18.3)	0.03	< 0.001
Sex, *n* (%)					
Male	197,611 (46.5)	17,892 (46.0)	179,719 (46.6)	0.00	0.026
Female	227,134 (53.5)	21,014 (54.0)	206,120 (53.4)		
Race, *n* (%)					
American Indian/Alaska Native/Native Hawaiian	787 (0.2)	161 (0.4)	626 (0.2)	0.40	< 0.001
Asian	8006 (1.9)	2757 (7.1)	5249 (1.4)		
Asian Indian	4602 (1.1)	1318 (3.5)	3284 (0.9)		
Black	68,660 (16.2)	1372 (3.6)	67,288 (17.5)		
Other[b]	35,680 (8.4)	15,087 (38.8)	20,593 (5.3)		
Pacific Islander	531 (0.2)	126 (0.3)	405 (0.2)		
Unknown/Declined to answer	9651 (2.3)	2635 (6.9)	7016 (1.8)		
White	296,828 (69.9)	15,450 (39.8)	281,378 (72.9)		
Ethnicity, *n* (%)					
Yes, Spanish/Hispanic/Latino	48,502 (11.5)	20,735 (53.3)	27,767 (7.2)	0.42	< 0.001
Not Spanish/Hispanic/Latino	348,476 (82.0)	16,003 (41.1)	332,473 (86.2)		
Unknown/Unavailable	27,767 (6.5)	2168 (5.6)	25,599 (6.6)		
Primary insurance status, *n* (%)					
Private/Employee-sponsored	170,317 (40.1)	15,640 (40.2)	154,677 (40.1)	0.11	< 0.001
Medicaid	24,763 (5.8)	3447 (8.9)	21,316 (5.5)		
Medicare	218,932 (51.5)	16,890 (43.4)	202,042 (52.4)		
Self-pay	10,733 (2.6)	2929 (7.5)	7804 (2.0)		
Charlson Comorbidity Index					
0	132,588 (31.2)	11,878 (30.5)	120,710 (31.3)	0.01	< 0.001
1	103,071 (24.3)	9213 (23.7)	93,858 (24.3)		
2	189,086 (44.5)	17,815 (45.8)	171,271 (44.4)		
Outcome					
7-day readmission[c], *n* (%)	16,803 (4.1)	1768 (4.5)	15,035 (3.9)	0.01	< 0.001

Abbreviations: LEP, limited English proficiency; SD, standard deviation.
[a]Sample excludes patients with length of stay > 60 days, behavioral health, and labor and delivery primary diagnoses.
[b]Other represents a race other than those available NJDDCS data dictionary.
[c]Excluded patients discharged December 26th or later, died in hospital, were discharged to hospice, or left against medical advice; Effect sizes were calculated as Cohen's D effects for mean differences, Phi (φ) effects when comparing two groups; and Cramer's V statistics for comparing more than 3 groups.

study sample. Patients with an indicated language preference other than English were classified as having LEP (Duraiswamy et al. 2022; Lopez et al. 2023; Santana et al. 2023).

Nurses were included ($n = 2839$) if they were actively licensed registered nurses in New Jersey and if they indicated on the RN4CAST-US survey that their primary job was in a hospital setting. Adult nonfederal, general acute care hospitals located in New Jersey with at least three nurse respondents to the RN4CAST-US survey reporting about the work environment were included ($n = 68$). The decision to use three nurses was supported by a moderate intraclass correlation of 0.7 and previous literature using 2016 RN4CAST data and comparable nurse respondent minimums (Lake et al. 2020; Muir et al. 2023). There was an average of 37 nurse respondents per hospital (range 3–129 nurse respondents per hospital). The final hospital sample represented 94.4% of general acute care hospitals in NJ.

2.3 | Variables

2.3.1 | Predictors

2.3.1.1 | Nurse Work Environment. Nurses reported on the quality of the nurse work environment using the Practice Environment Scale of the Nursing Work Index (PES-NWI), a

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31-item, National Quality Forum endorsed tool (National Quality Forum 2022). The PES-NWI contains five subscales: (1) nurse participation in hospital affairs, (2) nursing foundations for quality of care, (3) nurse manager ability, (4) staffing and resource adequacy, and (5) collegial nurse-physician relations (Lake 2002; Zangaro and Jones 2019). The survey items are measured on a 4-point Likert scale from strongly agree to strongly disagree. Nurses' reports of their work environments were combined for each subscale to form a hospital-level composite measure, calculated as the average of the five subscale scores. Each hospital-level nurse work environment score was then categorized as "poor" if the hospital score was in the bottom 25th percentile or "favorable" if in the top 75th percentile of scores. The definition of a poor work environment is consistent with decades of research using the PES-NWI instrument (Muir, McHugh, et al. 2023; White et al. 2020).

2.3.2 | Outcome

2.3.2.1 | Readmission. Seven-day readmission was defined as a rehospitalization occurring within 7-days of the discharge date from the initial hospitalization date. The odds of a 7-day readmission for LEP patients were compared to patients without LEP to assess the disparity in readmission between patients with and without LEP.

2.3.3 | Covariates

Patient covariates, provided by the NJDDCS data, included age, sex, comorbidities (measured by the Charlson Comorbidity Index, which categorizes ICD-10 patient diagnosis codes and assigns weights to comorbidities based on the associated risk of death within 1 year) (Charlson et al. 1987), diagnostic-related group (used by Medicare to categorize hospitalization costs) for risk adjustment, race, ethnicity, and insurance status. Hospital covariates, provided by the AHA data, included bed size, teaching status, and technology status. Bed size was defined as the number of beds in the hospital and was grouped as small (≤ 100 beds), medium (100–250 beds), and large (≥ 250 beds). Teaching status was determined by the ratio of medical residents or fellows per hospital bed. nonteaching hospitals have no medical residents/fellows. Minor teaching hospitals have a $< 1{:}4$ ratio. Major teaching hospitals have a $\geq 1{:}4$ ratio. High-technology status hospitals were defined as those with the ability to perform open-heart surgery or major organ transplantation.

2.4 | Statistical Analyses

All analyses were conducted using Stata 17 (*StataCorp* n.d.). Based on a statistical power analysis performed in PASS, our sample size was able to detect an odds ratio as low as 1.06 (80% power, $\alpha = 0.05$) (PASS 2022 Power Analysis and Sample Size Software, 2022 Power Analysis and Sample Size Software 2022). Descriptive statistics were performed (e.g., percentages, means, standard deviations) to examine the characteristics of the patient sample, comparing the total patient sample to those with and without LEP. Characteristics of the hospital sample and the quality of nurse work environments were described. *T*-tests and Chi-square tests were used to assess for significant differences between patients with and without LEP.

Sequential multivariable logistic regression models were performed in a series of unadjusted and adjusted models to determine the odds of a 7-day readmission for patients with and

TABLE 2 | Distribution of patient sample across hospital characteristics.

Characteristic	All ($n = 424{,}745$)	LEP ($n = 38{,}906$)	Non-LEP ($n = 385{,}839$)	Effect Size	*p* value
Work environment, *n* (%)					
Poor	80,639 (19.0)	9710 (25.0)	70,929 (18.4)	0.05	< 0.001
Favorable	344,106 (81.0)	29,196 (75.0)	314,910 (81.6)		
Bed size, *n* (%)					
< =100 Beds	4490 (1.1)	130 (0.3)	4360 (1.1)	0.02	< 0.001
101–250 Beds	96,926 (22.8)	8686 (22.3)	88,240 (22.9)		
> =250 Beds	323,329 (76.1)	30,090 (77.4)	293,239 (76.0)		
Teaching status[a], *n* (%)					
Nonteaching	141,407 (33.3)	8313 (21.4)	133,094 (34.5)	0.08	< 0.001
Minor teaching	178,537 (42.0)	19,775 (50.8)	158,762 (41.2)		
Major teaching	104,801 (24.7)	10,818 (27.8)	93,983 (24.3)		
Technology status[b], *n* (%)					
Not high technology	262,819 (61.9)	23,379 (60.1)	239,440 (62.1)	0.01	< 0.001
High technology	161,926 (38.1)	15,527 (39.9)	146,399 (37.9)		

Abbreviations: LEP, limited English proficiency; SD, standard deviation.
[a]Non-teaching: no medical residents/fellows, minor teaching: ratio of < 1:4 medical residents/fellows, major teaching: ratio of > = 1:4 medical residents/fellows.
[b]Not high-tech: do not have the ability to perform open-heart surgery or major organ transplantation, high-tech: have the capacity to perform open-heart surgery or major organ transplantation.

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TABLE 3 | Sequential multilevel logistic regression models of LEP disparities in 7-day readmissions in favorable vs poor nurse work environments.

	Model 1: Unadjusted	Model 2: Adjusted for patient characteristics	Model 3: Adjusted for patient and hospital characteristics	Model 4: Adjusted for patient and hospital characteristics, and nurse work environment	Model 5: Fully adjusted with interaction of LEP and nurse work environment
	OR [95% CI]	OR [95% CI]	OR [95% CI]	OR [95% CI]	OR [95% CI]
LEP[a]	1.17* [1.07–1.27]	1.33** [1.20–1.49]	1.33** [1.19–1.47]	1.32** [1.19–1.47]	1.51* [1.28–1.79]
AUC[b]	0.51	0.65	0.65	0.65	0.65
Nurse Work Environment[c]				0.99 [0.94–1.04]	1.01 [0.95–1.07]
Interaction (LEP × Nurse Work Environment)					0.83* [0.70–0.99]

Abbreviations: CI, confidence interval; OR, odds ratio.
[a]Reference group for LEP is non-LEP.
[b]AUC = Area under the ROC curve.
[c]The nurse work environment is a binary variable (poor vs. not poor), with poor as the referent group in the regression model; Patient characteristics include age, sex, Charlson Comorbidity Index, diagnostic-related groups (DRG), race, ethnicity, insurance; Hospital characteristics include teaching status, technology status, and bed size; *$p < 0.05$; **$p < 0.001$; Logistic regression accounted for clustering of patients within hospitals.

without LEP. An interaction between the nurse work environment and patients' LEP status on 7-day readmissions was also performed to understand whether and to what extent there were differential impacts of the nurse work environment for patients with LEP.

Five sequential models were constructed to isolate the individual and combined impacts of patient characteristics, hospital characteristics, and the nurse work environment on readmission disparities. First, an unadjusted model examined the bivariate association between 7-day readmissions and LEP status. Adjusted models accounted for patient characteristics, patient and hospital characteristics, patient and hospital characteristics along with the nurse work environment, and patient and hospital characteristics and an interaction term between patient LEP status and the hospital nurse work environment.

3 | Results

The overall patient sample was 424,745 patients, 9.2% of which were classified as having LEP ($n = 38{,}906$). Patient characteristics and readmissions in the overall patient sample and among patients with and without LEP are presented in Table 1.

Despite being younger on average (63.4 vs 64 years old, $p < 0.001$), patients with LEP were more likely to be readmitted (4.5% vs 3.9%, $p < 0.001$) and had more comorbidities as evidenced by 45.8% of patients with LEP having a Charlson Comorbidity Index of 2, as compared to 44.4% of patients without LEP ($p < 0.001$). Patients with LEP were also less likely to be White (39.7% vs 72.9%, $p < 0.001$) and more likely to be insured by Medicaid (8.9% vs 5.5%) or uninsured (7.5% vs 2%, $p < 0.001$) as compared to patients without LEP. Relative to patients without LEP, patients with LEP patients were more likely to receive care in hospitals with poor nurse work environments (25% vs 18.4%, $p < 0.001$), within larger teaching institutions (27.8% vs 24.4%, $p < 0.001$), and without high technology capabilities (39.9% vs. 37.9%, $p < 0.001$) (Table 2). Importantly, the effect sizes of these differences between patients with and without LEP were small, other than the effect sizes for differences among race and ethnicity (Tables 1 and 2).

In the unadjusted model, patients with LEP have 17% higher odds of a 7-day readmission, as compared to patients without LEP (OR 1.17, 95% CI [1.07–1.27]) (Table 3). After adjusting for patient and hospital characteristics, patients with LEP have 33% higher odds of a 7-day readmission, as compared to patients without LEP (OR 1.33, 95% CI [1.19–1.47]). When accounting for the nurse work environment, the LEP readmission disparity was slightly attenuated, although still statistically significant (OR 1.32, 95% CI [1.19–1.47]). The direct effect of the nurse work environment on 7-day readmissions was not statistically significant (OR 0.99, 95% CI [0.94–1.04]).

The final model in Table 3 represents the fully adjusted model and includes an interaction term between patient LEP status and the hospital nurse work environment. The interaction term was statistically significant (OR = 0.83, 95% CI [0.70–0.99]), indicating that patients with LEP experienced lower odds of a 7-day readmission when cared for in hospitals with favorable

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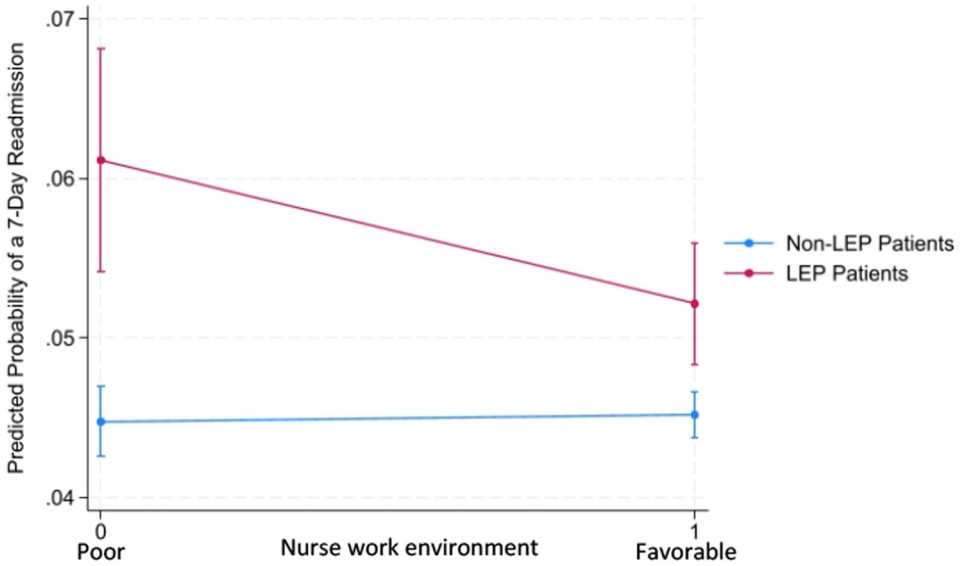

FIGURE 1 | Predicted probability of a 7-day readmission for patients with and without limited English proficiency in poor versus favorable nurse work environments. *Note:* LEP = Limited English proficiency. The error line represents a confidence interval. [Color figure can be viewed at wileyonlinelibrary.com]

versus poor nurse work environments, compared to patients without LEP. Patients with LEP experienced a 16% decrease ($0.83 \times 1.01 = 0.84$) in odds of a 7-day readmission when in favorable nurse work environments.

A margins plot displays the predicted probabilities of a 7-day readmission between patients with and without LEP across hospitals with poor and favorable environments (Figure 1). Notably, the disparity between readmission odds for patients with and without LEP is considerably narrower in favorable compared to poor nurse work environments.

4 | Discussion

With a diverse and growing population of patients with LEP in the United States (Betancourt et al. 2012), our study provides novel evidence that improvements to nurses' work environments can simultaneously address readmission disparities among patients with LEP. Importantly, our study shows that the higher odds of readmission among patients with language barriers are reduced in favorable work environments.

There are signals in the literature highlighting which factors of the nurse work environment may be particularly relevant to nurses caring for patients with LEP. For example, patients with LEP experience barriers to accessing healthcare (e.g., difficulties identifying language-concordant primary care providers covered by their insurance) which may be better addressed when nurses are afforded the additional time needed to acquire and use interpreters (Coleman and Angosta 2017; Gerchow et al. 2020; Squires et al. 2019).

The interventions that nurses most commonly identify as preferred solutions to enhance their ability to deliver high-quality and safe patient care are those that enable them to spend more time directly with patients (Aiken et al. 2023). Safe staffing (e.g., lower patient-to-nurse staffing ratios) is one solution to allow nurses to spend more time with patients. As such, favorable environments characterized by sufficient staffing, easily accessible resources, and supportive management may allow nurses to better meet the unique needs of patients with LEP. For example, nurses working in more favorable work environments may be more likely to acquire language services, establish connections to community-based resources, and address any barriers to effective care delivery to leadership than nurses working in hospitals with lower-quality nurse work environments (characterized by poor staffing and minimal resources) (Gerchow et al. 2020; Squires et al. 2019). In addition, although hospitals are legally required to provide language services, research indicates that the quality, accessibility, and utilization of these services vary significantly across different hospitals. (Jones 2008; Starr and Wallace 2009; Tam et al. 2020). Ensuring consistency and quality in interpreter services across hospitals is crucial for providing equitable, safe, and effective healthcare to all patients, regardless of their language proficiency (Heath et al. 2023). Therefore, it is important to consider how both the quality of the language services and the context of care—nurses' work environment—which enables accessibility and usability, may be especially consequential to health outcomes for patients with LEP.

Our findings expand the growing body of literature surrounding the systemic discrimination that patients with LEP experience while navigating the U.S. healthcare system (Escobedo et al. 2023; Gulati and Hur 2022; Yeheskel and Rawal 2019), as exhibited in this study by their increased likelihood to receive care in hospitals with poorer nurse work environments. The success of policies such as Section 1557 of the ACA–which aims to advance health equity goals–may be contingent on nurses having favorable nurse work environments to be able to fulfill

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policy requirements. The expanded ACA provisions related to language access may have impacted LEP care delivery and outcomes; however, a follow-up study using more recent NJ data is a crucial next step to assess whether the policy has been effective in reducing outcome disparities for patients with LEP.

4.1 | Limitations

The cross-sectional design of the study does not allow for the determination of causality, although longitudinal studies examining the relationship between hospital nursing resources and various patient outcomes have shown results similar to those of cross-sectional studies.(Kutney-Lee et al. 2015; Lasater et al. 2021; Sloane et al. 2018). Although the study data is from 2016 and is limited to a single state (NJ), it provides baseline results for a future policy examination of Section 1557 of the ACA. NJ is one of the most linguistically diverse states in the United States (U.S. Census Bureau n.d.). In addition, very few language disparities studies have examined outcomes across hospital systems, which adds a novel organizational perspective to the longstanding clinical challenge for how to provide high-quality and safe care to patients with LEP.

5 | Conclusions

Reducing inequities and disparities in hospital outcomes is critical for ensuring value-based care for all persons in the United States, regardless of their ability to communicate in English. This study adds to the evidence that the quality of the work environments of hospital-based nurses is a factor that further contributes to inequities and disparities in outcomes. Interventions that focus on improving the quality of work environments of nurses, therefore, have the potential to reduce inequities and disparities in outcomes like those seen amongst patients with LEP in this study.

Author Contributions

All co-authors have both contributed substantially to the paper, have approved the version being submitted, and have met the ICMJE expectations for authorship.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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